NOT MEASUREMENT SENSITIVE

MIL-M-83827
13 August 1992

MILITARY SPECIFICATION
MAGNET SWEEPER, ELECTROMAGNETIC TYPE, WHEEL MOUNTED, 96 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a two-wheeled self-contained, weatherproof, electromagnetic road sweeper with a gasoline or diesel engine driven generator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
MILITARY
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.

STANDARDS
FEDERAL
FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595 - Colors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins Air Logistics Center, ATTN: WR-ALC/LVRG, Robins AFB GA 31098-5345, by using the self-addressed Standarization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A FSC-3830

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MILITARY

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-970	- Standards and Specifications, Order of Preference for the Selection of.
MIL-STD-831	- Test Reports, Preparation of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-965	- Parts Control Program.
MIL-STD-1556	- Government/Industry Data Exchange Program Contractor Participation Requirements.
MS51336	- Lunette-Coupler, Drawbar, Ring.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DEPARTMENT OF DEFENSE (DOD)
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093).

US AIR FORCE (USAF)

Drawing 8838302	- Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicles and Support Equipment.

(Copies of this drawing required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 *Non-Government publications*. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Welding Qualifications - Section 9-86

(Application for copies should be addressed to American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York NY 10017.)

AMERICAN WELDING SOCIETY (AWS)
Welding Qualifications - B2.1-84

(Application for copies should be addressed to American Welding Society,

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2501 N.W. Seventh Street, Miami FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
J534 - Lubrication Fittings.
J1175 - Bystander Sound Level Measurement Procedure for Small Engine Powered Equipment.

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities, and services not specified herein shall be selected in accordance with MIL-STD-970.

3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as otherwise specified, wood shall not be used.

3.1.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3 Design and construction. The sweeper shall be a self contained unit complete with magnet, gasoline engine, generator, tow bar, parking stand, wheels, and tires. The sweeper shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations, and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair, and servicing. Its design shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, and replacement.

3.1.3.1 Commercial product. The sweeper and components must be the manufacturer's standard commercial product (see 6.6.1), except as modified to meet the requirements of this specification. Upon request of the contracting officer, the offeror must submit evidence to establish commerciality of the sweeper and furnish names and addresses of non-Government sources that have acquired the sweeper at least a year prior to the solicitation opening date.

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3.1.3.2 Maintainability. The sweeper shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources, and maintaining the sweeper. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing, and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

3.1.3.3 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except as provided in commercial (see 6.6.1) components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter. All fasteners of Grade 5 or higher must meet the quality assurance requirements of US Air Force Drawing 8838302.

3.1.3.4 Servicing provisions. Drains, grease fittings, lines, and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.3.5 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.3.6 Component protection. All space in which work is performed during operation, service, and maintenance of the sweeper shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.4 Temperature conditions. The sweeper shall be capable of continual operation at any ambient temperature from -20°F to +125°F and long term storage at any ambient temperature from -60°F to +160°F when serviced in accordance with the supplier's recommendations.

3.1.5 Sound level. The exterior sound level of the sweeper shall not exceed 84 dB(A) when measured in accordance with SAE J1175.

3.2 Performance.

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3.2.1 Sweeping pickup capability. At a speed of not less than 5 miles per hour (mph) and with a sweeping clearance of 4 inches, the sweeper shall be capable of picking up not less than 99 percent of 50 pounds of miscellaneous small steel items such as washers, nuts, wire and flattened cans that have been uniformly spread on the ground beneath the magnet assembly pickup area. At the manufacturer's option, 50 pounds of size 4d common steel nails may be used instead of the miscellaneous small steel items. When the magnet is de-energized, not less than 99 percent of the pickup material shall fall to the ground

3.2.2 Towing ability. The sweeper shall be capable of being safely towed, in the travel mode, over improved roads at a speed of not less than 30 mph and over unimproved roads at a speed of not less than 10 mph. The sweeper shall follow in the path of the towing vehicle without shifting or swerving more than 3 inches from side to side. The sweeper shall trail the towing vehicle without cramping or sideslipping when prime mover is turning at its minimum radius.

3.3 Components.

3.3.1 Dimensions. The sweeper shall have a wheel track width of not less than 108 inches, an overall height of not less than 35 inches, and an overall length of not more than 80 inches.

3.3.2 Wheels and tires.

3.3.2.1 Wheels. The wheels shall be mounted on hardened axles with tapered roller bearings. When the tires are properly inflated and under normal load, the wheel centers shall ride a minimum of 9.7 inches above the ground plane.

3.3.2.2 Tires. Tires shall be of the tube or tubeless type, high flotation and shall conform to all Tire and Rim Association recommendations for this application. The tires shall have individual rated load carrying capacities equal to the maximum individual tire loading imposed by the operation. The tires shall be capable of operating on a hard paved surface or in loose sand or dirt.

3.3.3 Generator set. The generator set shall be a close coupled type with the generator directly connected to the engine crankshaft in a manner that shall transmit the engine torque smoothly, and produce no injurious stresses to the shafts. The set shall be provided with a common base of substantial construction to prevent misalingment, and shall be secured to the sweeper by not less than four bolts. The mounting shall effectively cushion the generator set to prevent vibrations while the sweeper is being used or towed. A control switch shall be furnished to provide "on-off-drop" control. The electrical circuitry shall provide a reverse current capability to reverse the magnetic field for complete release of all residual magnetism.

3.3.3.1 Engine. Unless otherwise specified, the engine shall be a gasoline driven, 4 stroke cycle, air cooled, heavy-duty, industrial type, with a magneto ignition system, and a pressure, splash, or spray lubrication system. When specified (see 6.2), the engine shall be a diesel driven industrial type, 2 or 4 stroke cycle, air cooled. The fuel shall be suitably filtered, and an oil bath or dry type air cleaner shall be furnished. The exhaust system shall direct exhaust gases away from the operating controls. The engine speed shall be maintained within $\pm$ 5 percent of rated speed for constant loads or gradual load

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changes. There shall be no sustained periodic speed oscillations under conditions of constant load operation, and following a sudden change in load, the engine speed shall return to steady state conditions. The engine speed and power shall be sufficient for the intended purpose. The engine shall have a manual starting system. A fuel tank shall be provided with a fuel capacity sufficient for not less than 8 hours continuous operation at full load.

3.3.3.2 Generator. The generator shall be self excited and shall be provided with inherent voltage regulation that shall maintain the generator terminal voltage within ± 5 percent of rated voltage for any constant load between no load and full load. The generator shall have a continuous duty rating of not less than 3000 watts at 125 volts DC. The generator shall be provided with a double sealed, prelubricated bearing at the outboard end of the rotor shaft. The generator enclosure shall be of a drip proof design, and shall protect the generator from dust and moisture. The rated temperature rise of the generator shall be not greater than 50°C at an ambient temperature of 40°C.

3.3.4 Electromagnet. The sweeper electromagnet shall be of high flux intensity with heavy section conductors. Coils shall be layer wound and vacuum impregnated with high grade insulating varnish. Any cavities between the coil and steel magnet structure shall be filled with a high dielectric, semirigid compound of a high thermal conductivity, providing maximum resistance to coil damage through mechanical shock and rough handling. The magnet shall be enclosed in a weatherproof housing such that additional framing for the carrying mechanism is not required. There shall be no magnetic "dead spots" on the pickup surface of the housing. The magnet shall have an effective sweeping width of 96 inches and a depth of 16 inches at a nominal ground clearance of 4 1/2 inches and shall be adjustable between, 4 and 7 inches above the ground. Such adjustment shall not affect the normal horizontal plane of the magnet. The strength of the magnetic field, measured along the centerline of the face of the magnet, shall be 1080 Gauss minimum at a distance of one inch from the magnet face and 240 Gauss minimum at a distance of 8 inches from the magnet face.

3.3.5 Tow bar. The tow bar shall have a lunette type towing ring in accordance with MS51336. The tow bar shall be constructed from material structurally adequate for the weight drawn. The sweeper shall be designed with its center of gravity forward of the wheel axle center line, thereby placing a weight load on the towing ring. The weight on the towing ring shall not be more than 100 pounds or less than 50 pounds. Provision shall be made for adjusting the height of the tow bar from 10 to 30 inches to match the height of the towing vehicle hitch. Such adjustment shall not disturb the horizontal plane of the magnet.

3.3.5.1 Safety chains. The sweeper shall be provided with safety chains to attach to the towing vehicle. The chains shall be not less than 30 inches long and fitted with grabhooks. The safety chains shall be arranged and connected such that they will prevent the tow bar from dropping to the ground in the event of hitch failure and so that there is no excess slack in the chains other than what is necessary to permit proper turning. Each chain shall have an ultimate strength at least equal to the gross weight of the load being towed.

3.3.5.2 Parking stand. A parking stand, with a landing pad, shall be furnished with the tow bar. The parking stand shall fold into the tow bar when the sweeper is in use. The parking stand shall be bolted to the tow bar and shall lock in both the folded and supporting positions.

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3.4 Finish and protective coatings.

3.4.1 Painting. The entire exterior surface of the sweeper shall be painted in accordance with the manufacturer's standard commercial (see 6.6.1) practice except that the color of the finish coat shall be Forest Green, Color Number 24052, of FED-STD-595.

3.4.1.2 Rustproofing. Unless otherwise specified, the entire sweeper shall be rustproofed in accordance with the manufacturer's standard commercial (see 6.6.1) practice. When specified (see 6.2), the entire sweeper shall be rustproofed, in accordance with the requirements for trailers and semitrailers in FED-STD-297. Stainless steel parts need not be rustproofed. The first article unit shall not be rustproofed until after approval of the first article test report.

3.5 Data plates and operating instructions. Data plates shall be in accordance with MIL-P-15024 Type B, Style II. The plates shall be securely attached to each sweeper with nonferrous screws or rivets of not less than 1/8 inch diameter, and shall be permanently and legibly marked in accordance with MIL-STD-130.

3.5.1 Instruction plate. The sweeper shall be equipped with an instruction plate(s), suitably located, describing any special or important procedures to be followed in operating or servicing the equipment.

3.5.2 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the sweeper.

b. Horizontal and vertical location of the center of gravity of the sweeper in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width, and height.

3.5.3 Identification plate. An identification data plate shall be provided. The plate shall contain at least the following information:

1. Nomenclature.

2. Manufacturer's model number.

3. Manufacturer's serial number.

4. Special characteristics; HP, RPM, etc.

5. Reg Number.

6. National Stock Number (NSN).

7. Manufacturer's name.

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8. Contract number.

9. Date of manufacture.

10. US.

3.6 Parts control. The contractor shall participate in the Government Industry Data Exchange Program (GIDEP) in accordance with MIL-STD-1556. In the design of the sweeper, the contractor shall ensure that the components selected have not been identified as a problem or unresolved deficiency by GIDEP. The contractor shall submit an Alert/Safe-Alert on a DD Form 1939 whenever an actual or potential problem is evident with respect to a part, component, material, process, safety condition or to test equipment. The contractor shall certify maximum use of off-the-shelf items in accordance with MIL-STD-965.

3.6.1 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.8 Fabrication.

3.8.1 Welders and welding.

3.8.1.1 Welders. All welders employed in the fabrication of the sweeper, except for commercial (see 6.6.1) components, shall be certified in accordance with the applicable requirements of the American Welding Society or the American Society of Mechanical Engineers.

3.8.1.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof and service loadings.

3.8.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.8.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of an approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the component.

3.8.4 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

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3.8.5 Cleaning. The sweeper shall be cleaned. Loose, spattered, or excess solder, welding slag, stray bolts, nuts and washers, rust, metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.9 Workmanship. The sweeper, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the sweeper conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the sweeper shall be classified as follows:

a. First article testing (see 4.4 and 6.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at the start of each test and then at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

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d. Ambient temperature.

e. Engine speed.

4.3.3 Preparation for tests. The sweeper, equipped with all specified equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the sweeper shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel or lubricant, under any condition except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between sweeper components.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering and braking.

h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

i. Overheating of the engine, generator, or any other sweeper component.

4.4 First article testing.

4.4.1 First article test sample. The first article test sample shall be identical to the items procured under the terms of the contract. Approval of the first article test report shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article tests. The first article tests shall consist of all tests listed under 4.6.

4.4.3 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All data collected, failures, maintenance performed, and other events recorded shall be identified by accumulated operation time, miles, cycles or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

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b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.

d. List of special tools and diagnostic equipment necessary to perform field level maintenance.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.5.1).

b. Random sample testing (see 4.5.2).

All defects identified shall be corrected prior to acceptance.

4.5.1 Individual testing. Each production sweeper shall be subjected to the tests specified in 4.6.1 and 4.6.2.

4.5.2 Random sample testing. When specified (see 6.2), any production sweeper may be selected by the procuring activity at any time to be subjected to all of the tests described in 4.6. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report.

4.5.2.1 Random sample test failure. When a sweeper fails to successfully complete any random sample test, the cause of the failure shall be determined, corrective action shall be taken, and the test shall be repeated to verify adequacy of the corrective action. Tests not involving the defective component(s) may be continued pending the investigation of the failure. Sweepers to be delivered pursuant to the contract shall be inspected for all defects identified during all random sample tests and may not be accepted prior to verification of adequacy of the corrective action for a random sample test failure.

4.6 Test methods.

4.6.1 Examination of product. The sweeper shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting and markings. For the first article and random sample sweepers this examination shall be accomplished by use of a checklist which lists each requirement of section 3 not validated by a test and the results of the examination.

4.6.2 Preliminary operation. Sweepers produced in fulfillment of the contract or order shall be completely assembled, adjusted, lubricated and otherwise serviced for operation. A run-in test shall be conducted for each sweeper before delivery in accordance with the manufacturer's recommendations. During this operational test, all controls, adjusting mechanisms and other accessories

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shall be operated sufficiently to ensure against restrictions or malfunctions. Failure to pass any phase of this test shall be cause for rejection.

4.6.3 Sweeper pickup and residual test. The sweeper shall be tested to determine conformance to 3.2.1. The debris to be picked up shall be distributed across the full width of the magnet's surface.

4.6.4 Road test. The sweeper shall be road tested by the contractor to determine conformance to 3.2.2. The sweeper, with all equipment specified herein, shall be towed for a minimum distance of 30 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. The sweeper shall be towed at speeds up to at least 30 mph on paved surfaces and up to at least 10 mph on unpaved surfaces. Upon completion of the road test, all equipment shall be thoroughly examined. The equipment shall show no evidence of damage.

4.6.5 Design verification.

4.6.5.1 Sound level test. The sweeper shall be tested in accordance with SAE J1175 to demonstrate conformance to 3.1.5. The sweeper shall be tested in a traveling mode in accordance with the procedures for miscellaneous equipment (paragraph 3.2.6 of SAE J1175).

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the terms of the contract.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The magnet sweepers specified herein are intended for general purpose use in clearing roads, parking areas, airstrips, taxiways and ramps of tramp iron; and in cleaning gunnery ranges and strafing pits of expended shells and shell casings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

(c) If a diesel driven engine is required (see 3.3.3.1).

(d) If rustproofing in accordance with FED-STD-297 is required (see 3.4.1.2).

(e) If random sample testing is required (see 4.5.2).

6.3 Consideration of data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item

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Requirements List (CDRL), incorporated into the contract. When the provisions of DAR FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
4.4.3	First Article Test Report	DI-NDTI-80809A
4.5.2	Random Sample Test Report	DI-NDTI-80809A

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed in DOD 5000.19L, Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Sweeper
Magnetic
Electromagnet
Towed

6.6 Definitions.

6.6.1 Commercial. A manufacturer's standard (product, process or service) having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities such as samples, prototypes or experimental units do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification items acquired for "Government end use" include items acquired for foreign military sales.

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MILITARY INTERESTS:

Custodians

Air Force - 99
Navy - YD

PREPARING ACTIVITY:

Air Force - 84

Project No. 3830-0170

FSC 3830

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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c. ADDRESS (Include Zip Code) ROBINS AFB, GA 31098-5609	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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